

SOFTWARE ENGINEERING

LAB-3:Component Modelling & Architectural Pattern Selection

NAME:PRARTHANA HERUR

SRN:PES2UG24CS367

SECTION:F

DATE:9-09-2026

System: Community Lost & Found Matching Network

Architecture Selected: Layered Architecture

Architecture Selection

We chose **Layered Architecture** for the Community Lost & Found Matching Network.

Reason 1 — Separation of Concerns

The system contains different responsibilities such as user interaction, report management, item matching, claim processing, administration, and data storage. Layered Architecture separates these responsibilities into Presentation, Business, and Data layers, making the system easier to understand and maintain.

Reason 2 — Maintainability

The major system functions can be implemented as separate business components such as Report Management, Matching Service, Claim Management, and Admin Management. Changes to one component can therefore be made with minimal impact on other parts of the system.

Security Advantage

The Business Layer can control access to sensitive claim information before it is retrieved or modified. This helps restrict private claim details to authorized users and Community Admins, addressing the system's security requirement.

Performance Benefit

The Matching Service can be optimized separately to efficiently search and compare lost and found reports using item details, location, and date. This supports the requirement that potential matches should be displayed within five seconds after report submission.

Component Identification:

Component	Layer	Responsibility
User Interface	Presentation	Provides the web/mobile interface and handles user interactions
Report Management	Business	Validates reports, creates reports, generates report IDs, and updates reports
Matching Service	Business	Finds potential matches by comparing item details, location, and date
Claim Management	Business	Handles claim submission and tracks claim status
Admin Management	Business	Handles report/claim review and approval or rejection
Database	Data	Stores system and report-related information
Notification Service	Data/External Service	Sends match, claim, and status notifications

Component Interfaces:

Interface	Components Connected	Purpose
Report Submission API	User Interface → Report Management	Sends lost/found report information
Match Request API	Report Management → Matching Service	Requests potential-match processing
Report Query	Matching Service → Database	Retrieves report data required for matching
Claim Submission API	User Interface → Claim Management	Sends item claim information
Claim Review Interface	Claim Management ↔ Admin Management	Allows administrators to review claims
Notification API	Claim/Admin Management → Notification Service	Sends claim, match, and status notifications

Component Interaction / Data Flow

The overall interaction between components is as follows:

User Interface → Report Management → Matching Service → Database

For claims:

User Interface → Claim Management → Admin Management

For notifications:

Claim Management / Admin Management → Notification Service

The Notification Service is used to communicate match results, claim updates, and status notifications.

UML Component Diagram for Community Lost & Found Matching Network

